Glossary of Key Terms

Plain-language meanings of the words used in all sessions, in alphabetical order

HOW TO USE IT
Ask students to build their own glossary during each session. A good test: can they explain the word without looking at the definition?

Terms A–Z

Term	Meaning	Session
Abiotic	The non-living parts of an ecosystem: sunlight, water, soil, air, temperature.	Ecology
Activation energy	The energy "push" a reaction needs to get started. Enzymes lower it.	Enzymes
Active site	The pocket on an enzyme where the substrate binds.	Enzymes
Agar	A jelly from seaweed used to make solid culture media. Few bacteria can digest it.	Culture
Allolactose	A form of lactose that binds the lac repressor and switches the lac operon on.	Day 2
Amino acid	The building block of proteins. 20 are used in living things.	Day 1
Annealing	The PCR step in which primers bind to their matching sequence on the template.	PCR
Anticodon	Three bases on a tRNA that pair with a codon on the mRNA.	Day 1
Antibiotic	A substance that kills bacteria or stops them growing.	Culture
Apex predator	A top predator with no natural enemies except humans.	Ecology
Autoclave	A machine that sterilises with steam under pressure (121 °C).	Culture
Autotroph	An organism that makes its own food (producer).	Ecology
Bacteriophage	A virus that infects bacteria.	Day 3
Base pair	Two bases joined across the DNA double helix: A–T and G–C.	Day 1
Bioaccumulation	Build-up of a substance in one organism over its lifetime.	Ecology
Biomagnification	A poison becoming more concentrated at each higher level of a food chain.	Ecology
Biomass	The mass of living matter, and the energy stored in it.	Ecology

Term	Meaning	Session
Biopesticide	A pesticide made from natural materials such as plants, bacteria or minerals.	Ecology
Biotic	The living parts of an ecosystem.	Ecology
Bt	*Bacillus thuringiensis*, a bacterium whose proteins kill certain insect larvae.	Ecology
cAMP	Cyclic AMP, a small molecule that rises when glucose is low and activates CAP.	Day 2
CAP	Catabolite activator protein, a "glucose sensor" that boosts lac transcription.	Day 2
Catalyst	A substance that speeds up a reaction without being used up.	Enzymes
cDNA	DNA made from an RNA template by reverse transcriptase.	PCR
Central dogma	The idea that information flows from DNA to RNA to protein.	Day 1
Codon	A group of three bases in mRNA that codes for one amino acid or a stop.	Day 1
Colony	A visible cluster of bacteria grown from a single cell (CFU).	Culture
Competent cells	Cells that can take up DNA from their surroundings.	Day 3
Complementary	Matching by base-pairing rules (A with T/U, G with C).	Day 1
Conjugation	Direct transfer of DNA (such as a plasmid) from one bacterium to another.	Day 3
Control	A set-up that lets you check that the experiment works or is not contaminated.	General
Ct value	The cycle at which a real-time PCR signal crosses the threshold; low Ct = more starting material.	PCR
Decomposer	A fungus or bacterium that breaks down dead matter and returns nutrients.	Ecology
Denaturation	Unfolding of a protein, or separation of DNA strands, usually by heat.	Enzymes, PCR
Detritivore	An animal that eats dead plant and animal remains (e.g. earthworm).	Ecology
DNA ligase	The enzyme that joins DNA pieces by sealing nicks in the backbone.	Day 1, Day 3
DNA polymerase	The enzyme that builds new DNA strands on a template.	Day 1, PCR
Ecosystem	A community of organisms together with their non-living environment.	Ecology
Enzyme	A biological catalyst, usually a protein.	Enzymes

Term	Meaning	Session
Essential amino acid	An amino acid the body cannot make and must get from food (9 for adults).	Day 1
Exon	A part of a gene that stays in the mature RNA.	Day 1
Fermentation	Getting energy from sugar without oxygen; yeast makes ethanol and CO_2.	Yeast
Food web	All the connected food chains of an ecosystem.	Ecology
Frameshift	A change in the way codons are read, caused by missing or extra bases.	Day 1
Gel electrophoresis	Separating DNA by size in an electric field using a gel.	PCR
Gene	A section of DNA that carries the instructions for one RNA or protein.	Day 1
Generation time	The time a cell takes to divide into two.	Culture
Genome	All the DNA of an organism.	Day 1
Heterotroph	An organism that gets food from other organisms.	Culture, Ecology
Intron	A part of a gene that is removed from the RNA by splicing.	Day 1
IPTG	A lab chemical that switches on the lac operon but cannot be broken down.	Day 2, Day 3
lac operon	A group of bacterial genes (*lacZ*, *lacY*, *lacA*) for using lactose.	Day 2
Lagging strand	The new DNA strand made in short Okazaki fragments during replication.	Day 1
Ligation	Joining of DNA fragments by DNA ligase.	Day 3
Medium (culture)	The food mixture on or in which microbes are grown.	Culture
mRNA	Messenger RNA, the copy of a gene that is read by the ribosome.	Day 1
Multiple cloning site (MCS)	A short stretch on a plasmid with many restriction sites, for inserting genes.	Day 3
Nucleoside	A sugar plus a base (no phosphate).	Day 1
Nucleotide	A sugar plus a base plus a phosphate; the building block of nucleic acids.	Day 1
Okazaki fragments	Short DNA pieces made on the lagging strand.	Day 1
Operator	The DNA site where a repressor binds to block transcription.	Day 2
Operon	A group of genes controlled together from one promoter.	Day 2

Term	Meaning	Session
Palindrome (DNA)	A sequence that reads the same 5′→3′ on both strands (e.g. GAATTC).	Day 3
PCR	Polymerase chain reaction: a method to copy a chosen piece of DNA many times.	PCR
Peptide bond	The bond that joins amino acids in a protein.	Day 1
Plasmid	A small circular DNA molecule that copies itself separately from the chromosome.	Day 3
Primer	A short piece of nucleic acid that gives polymerase a starting point.	Day 1, PCR
Promoter	The DNA site where RNA polymerase starts transcription.	Day 1, Day 2
Quorum sensing	Bacterial communication by signal molecules that shows how dense the population is.	Day 2
Recombinant DNA	DNA made by joining pieces from different sources.	Day 3
Replication	Copying of DNA before a cell divides.	Day 1
Repressor	A protein that binds DNA and blocks transcription.	Day 2
Restriction enzyme	An enzyme that cuts DNA at a specific short sequence.	Day 3
Reverse transcriptase	An enzyme that makes DNA from an RNA template.	PCR
Ribosome	The machine that reads mRNA and builds proteins.	Day 1
RNA polymerase	The enzyme that makes RNA from a DNA template.	Day 1
Selective medium	A medium containing something that stops unwanted microbes growing.	Culture
Splicing	Cutting out introns and joining exons in pre-mRNA.	Day 1
Sticky end	A short single-stranded overhang left by a restriction enzyme.	Day 3
Substrate	The molecule an enzyme acts on.	Enzymes
Taq polymerase	A heat-stable DNA polymerase from *Thermus aquaticus*, used in PCR.	PCR
Transcription	Making RNA from a DNA template.	Day 1
Transduction	Transfer of DNA into a cell by a virus.	Day 3
Transfection	Forcing nucleic acids into animal cells.	Day 3
Transformation	Uptake of DNA by a bacterial cell.	Day 3
Translation	Reading mRNA to make a protein.	Day 1
tRNA	Transfer RNA, the adaptor that carries an amino acid to the ribosome.	Day 1
Trophic level	A feeding level in a food chain.	Ecology

Term	Meaning	Session
Vector	A DNA carrier (such as a plasmid) used to move genes into a host.	Day 3
X-gal	A colourless substance that turns blue when cut by β-galactosidase.	Day 3